

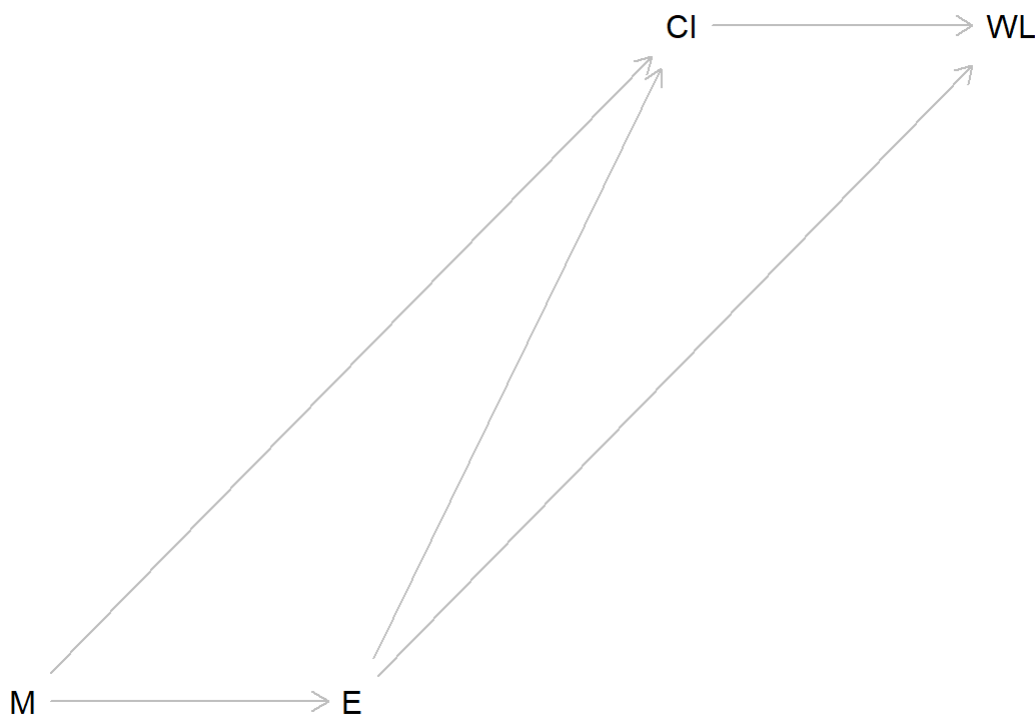
homework 1 causal inference

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problem 1: effect of exercise on weight loss

```
p1_dag <- dagitty("dag {
M -> E
M -> CI
E -> CI
E -> WL
CI -> WL
}")
coordinates(p1_dag) <- list(
  x = c(M = 0, E = 1.2, CI = 2.4, WL = 3.6),
  y = c(M = 1, E = 1, CI = 0, WL = 0)
)
plot(p1_dag)
```



dag: M -> E M -> CI E -> CI E -> WL CI -> WL

include in the model: CI

do not include: M

explanation: the researchers want to study the direct effect of regular exercise (E) on weight loss (WL). diet (CI) might also affect weight loss and is influenced by exercise. also motivation (M) could influence both E and CI.

because we want the direct effect, diet (CI) is a mediator on the path E -> CI -> WL, so we have to control for CI to block that path. M does not need to be in the model here, because once CI is included, the backdoor path through M is already blocked. i set all effect coefficients to 1 in the simulation.

```

sim_p1 = function(n=100) {
  M = rnorm(n)
  E <- rnorm(n, M)
  CI = rnorm(n, M + E)
  WL <- rnorm(n, E + CI)

  # correct: control for CI to get direct effect
  correct = coef(lm(WL ~ E + CI))['E']
  # wrong: only looking at E gives total-ish effect, not the direct one
  wrong = coef(lm(WL ~ E))['E']

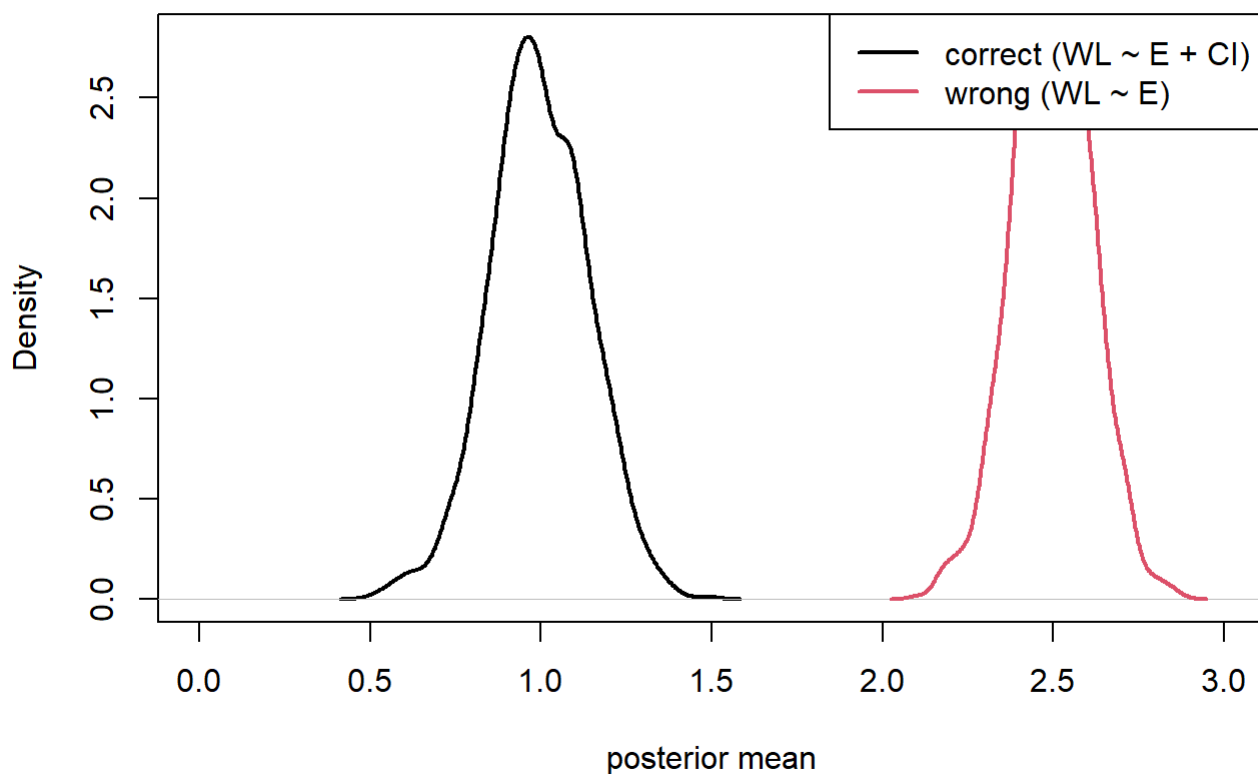
  return(c(correct, wrong))
}

res1 <- replicate(1000, sim_p1())

plot(density(res1[1,]), lwd=2, main="prob 1: direct effect E -> WL", xlab="posterior mean", x
lim=c(0, 3))
lines(density(res1[2,]), lwd=2, col=2)
legend("topright", legend=c("correct (WL ~ E + CI)", "wrong (WL ~ E)"), col=c(1, 2), lwd=2)

```

prob 1: direct effect E -> WL

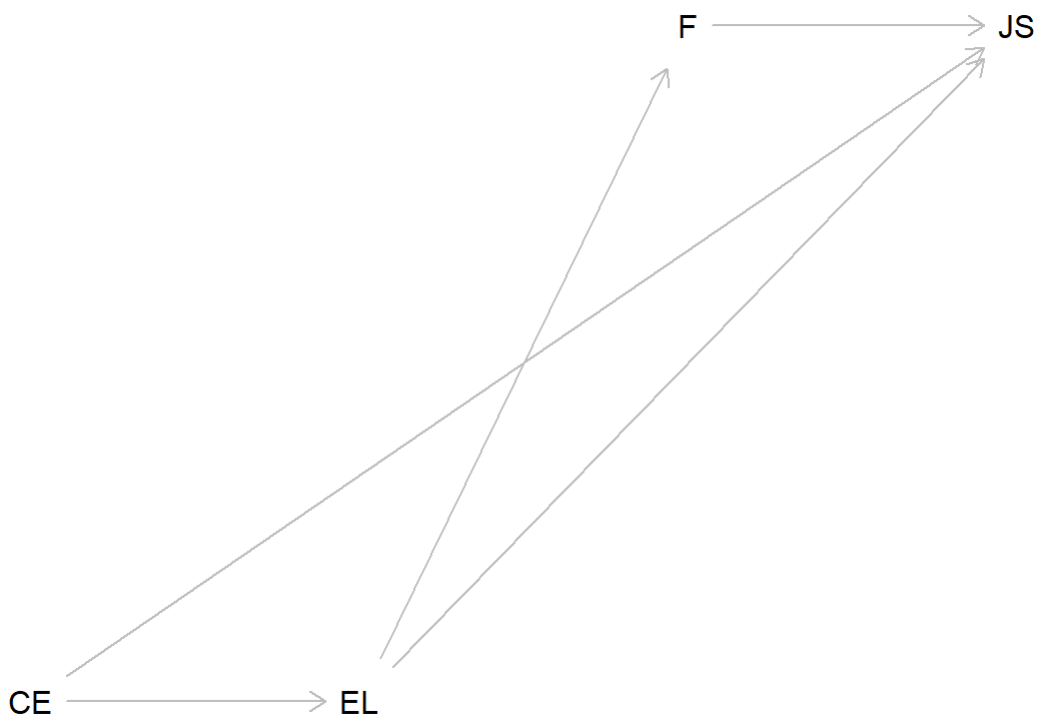


problem 2: education level and job satisfaction

```

p2_dag <- dagitty("dag {
CE -> EL
CE -> JS
EL -> F
F -> JS
EL -> JS
}")
coordinates(p2_dag) <- list(
  x = c(CE = 0, EL = 1.2, F = 2.4, JS = 3.6),
  y = c(CE = 1, EL = 1, F = 0, JS = 0)
)
plot(p2_dag)

```



dag: CE -> EL CE -> JS EL -> F F -> JS EL -> JS

include in the model: CE

do not include: F

explanation: here the study aims to investigate the direct effect of education level (EL) on job satisfaction (JS). the field of study (F) is influenced by EL and can also influence JS. cultural environments (CE) affect both EL and JS.

wait the task actually says we need to estimate the total effect of education level. so F is a mediator. if we control for F we block part of the total effect we want to measure, so we MUST NOT include F in the model. CE is a confounder so we have to include it to block the backdoor path.

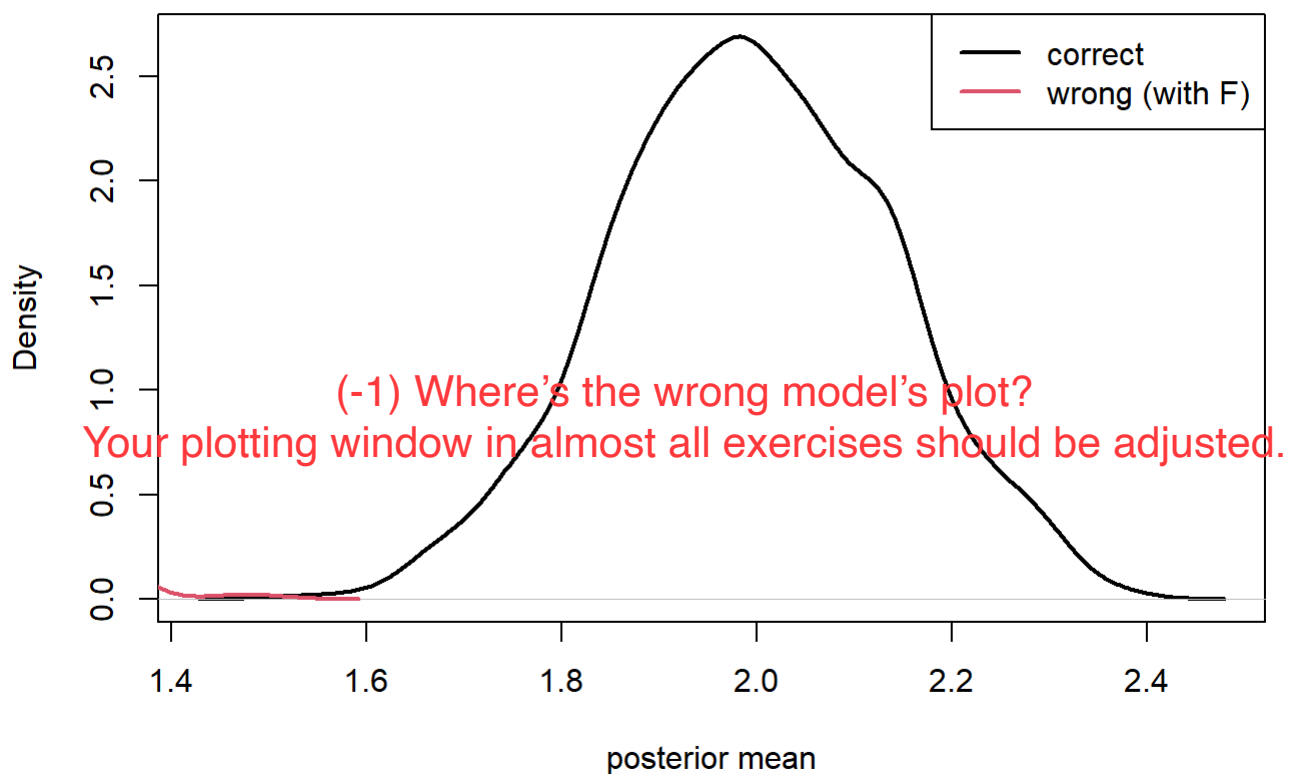
```
f2 = function(n=100) {
  CE <- rnorm(n)
  EL = rnorm(n, CE)
  F_var <- rnorm(n, EL)
  JS = rnorm(n, CE + F_var + EL)

  # correct: control CE only for total effect
  correct <- coef(lm(JS ~ EL + CE))['EL']
  # wrong: controlling for F blocks part of total effect
  wrong = coef(lm(JS ~ EL + CE + F_var))['EL']

  return(c(correct, wrong))
}

sim_2 = replicate(1000, f2())
plot(density(sim_2[1,]), lwd=2, main="prob 2: total effect EL -> JS", xlab="posterior mean")
lines(density(sim_2[2,]), lwd=2, col=2)
legend("topright", legend=c("correct", "wrong (with F)"), col=c(1, 2), lwd=2)
```

prob 2: total effect EL -> JS

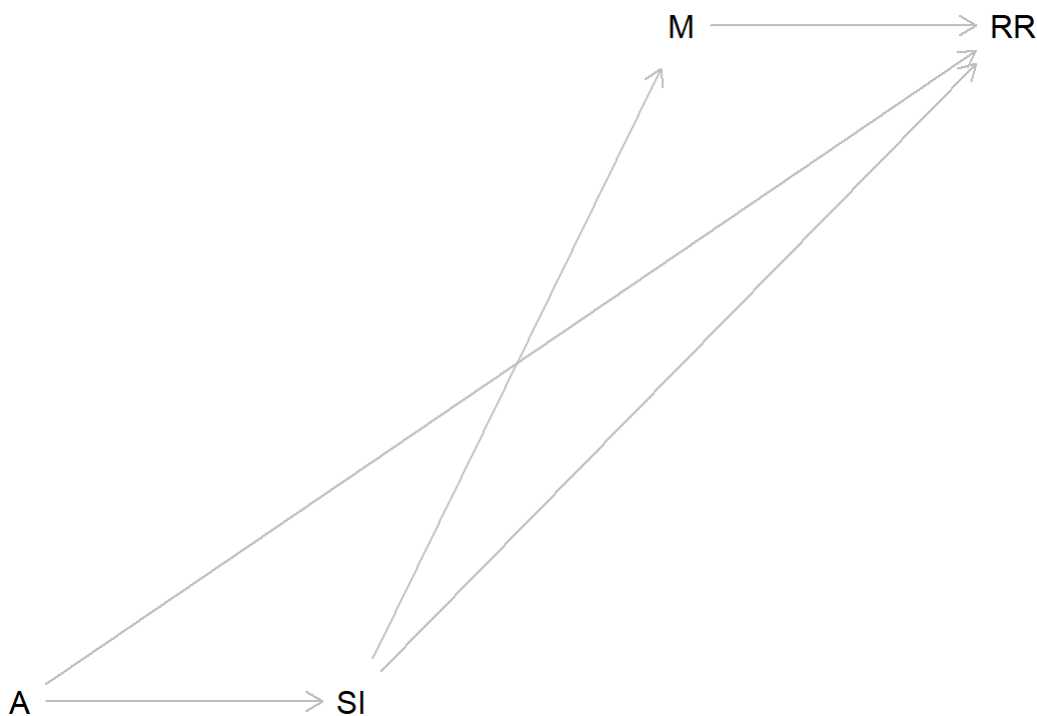


problem 3: medication use and patient recovery rate

```

p3_dag <- dagitty("dag {
A -> SI
A -> RR
SI -> M
SI -> RR
M -> RR
}")
coordinates(p3_dag) <- list(
  x = c(A = 0, SI = 1.2, M = 2.4, RR = 3.6),
  y = c(A = 1, SI = 1, M = 0, RR = 0)
)
plot(p3_dag)

```



dag: A -> SI A -> RR SI -> M SI -> RR M -> RR

include in the model: SI

optional extra control for precision: A

explanation: aim is to examine the effect of a new medication (M) on recovery rate (RR). severity of illness (SI) influences M and RR. age (A) influence RR and SI.

to asses the direct effect we must control for SI because its the actual confounder of M and RR.

part 2 asks what happens if you control for age in the model. age is not needed for identification once SI is already in the model, because SI already blocks the backdoor path. but if we control for A in addition to SI, the estimate usually gets more precise (distribution gets narrower), because age still explains some extra variance in RR. so it changes precision more than the estimand.

```

func_3 <- function(n=100) {
  A = rnorm(n)
  SI <- rnorm(n, A)
  M = rnorm(n, SI)
  RR <- rnorm(n, SI + A + M)

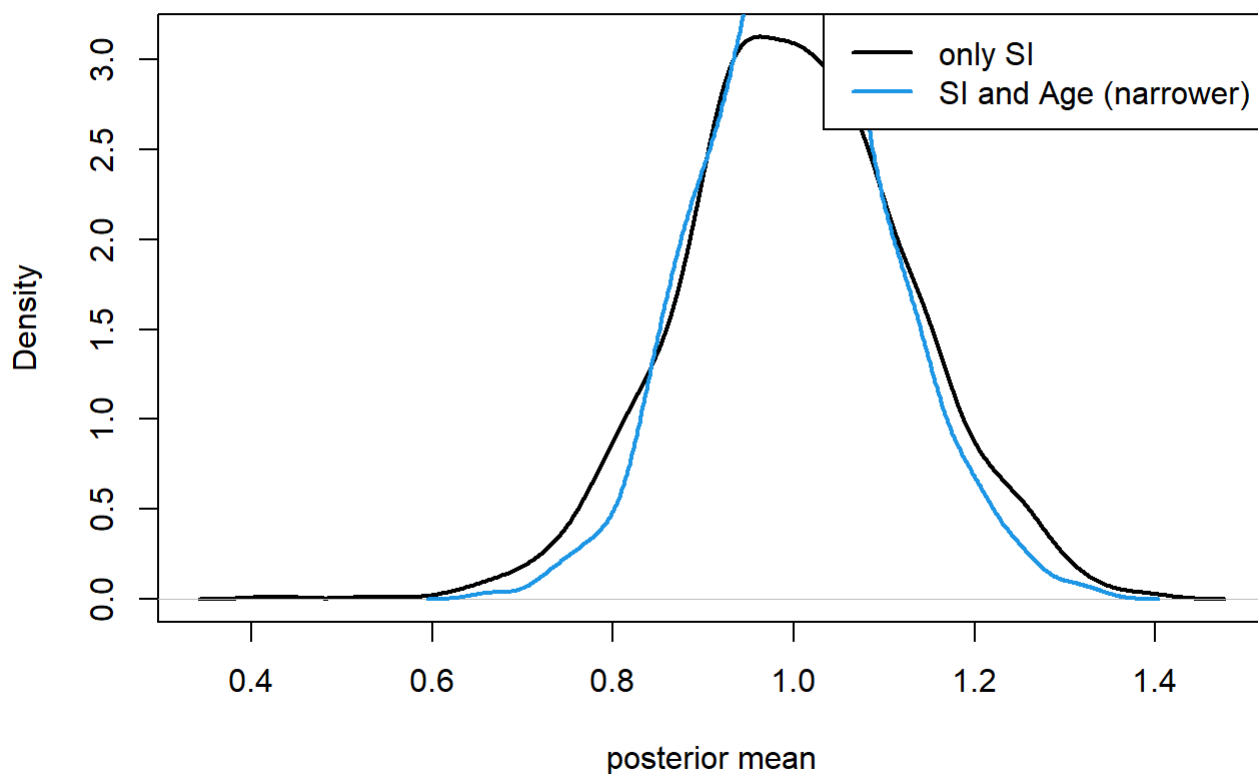
  # model with SI
  model_si <- coef(lm(RR ~ M + SI))['M']
  # model with SI and Age
  model_age = coef(lm(RR ~ M + SI + A))['M']

  return(c(model_si, model_age))
}

sim3 <- replicate(1000, func_3())
plot(density(sim3[1,]), lwd=2, main="prob 3: controlling SI vs SI+Age", xlab="posterior mean")
lines(density(sim3[2,]), lwd=2, col=4)
legend("topright", legend=c("only SI", "SI and Age (narrower)" ), col=c(1, 4), lwd=2)

```

prob 3: controlling SI vs SI+Age

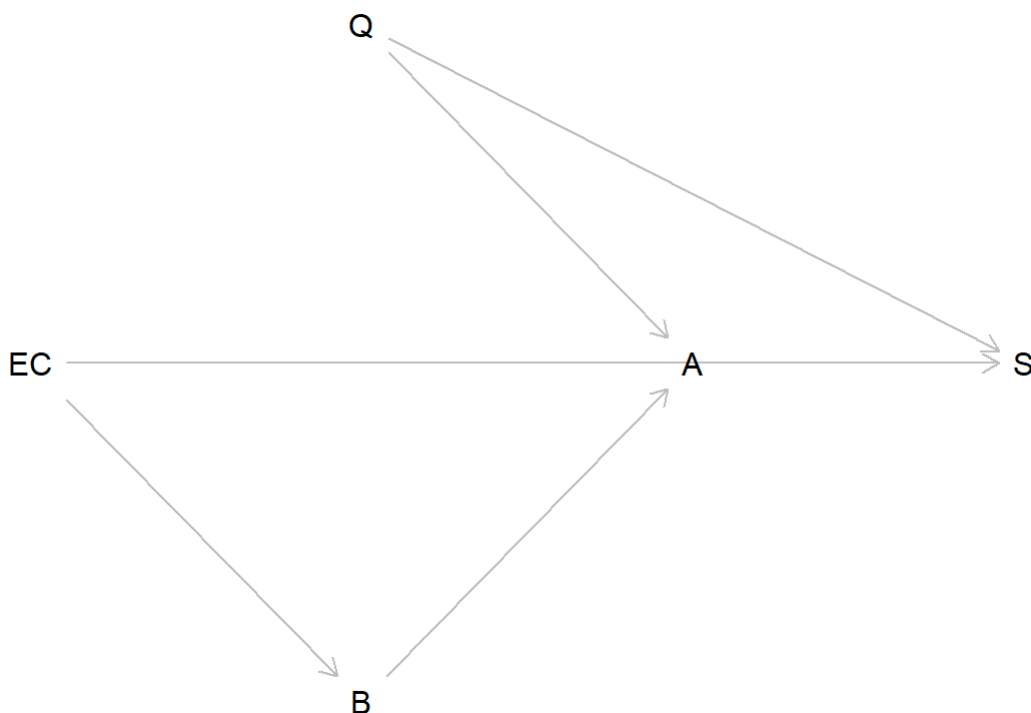


problem 4: advertising and sales

```

p4_dag <- dagitty("dag {
EC -> B
EC -> S
B -> A
Q -> S
Q -> A
A -> S
}")
coordinates(p4_dag) <- list(
  x = c(EC = 0, B = 1.2, Q = 1.2, A = 2.4, S = 3.6),
  y = c(EC = 1, B = 1.6, Q = 0.4, A = 1, S = 1)
)
plot(p4_dag)

```



dag: EC -> B EC -> S B -> A Q -> S Q -> A A -> S

include in the model: Q and EC
(or Q and B also works)

explanation: we want to evaluate the effect of advertising (A) on sales (S). economic climate (EC) influence budget (B) and S. quality (Q) affect S and A.

to estimate the total effect of A on S we have to close all backdoors. the paths are $A \leftarrow Q \rightarrow S$ and $A \leftarrow B \leftarrow EC \rightarrow S$. so we must control for quality (Q). and to block the economic path we can control for budget (B) or economic climate (EC). i will control for EC in the code.

```

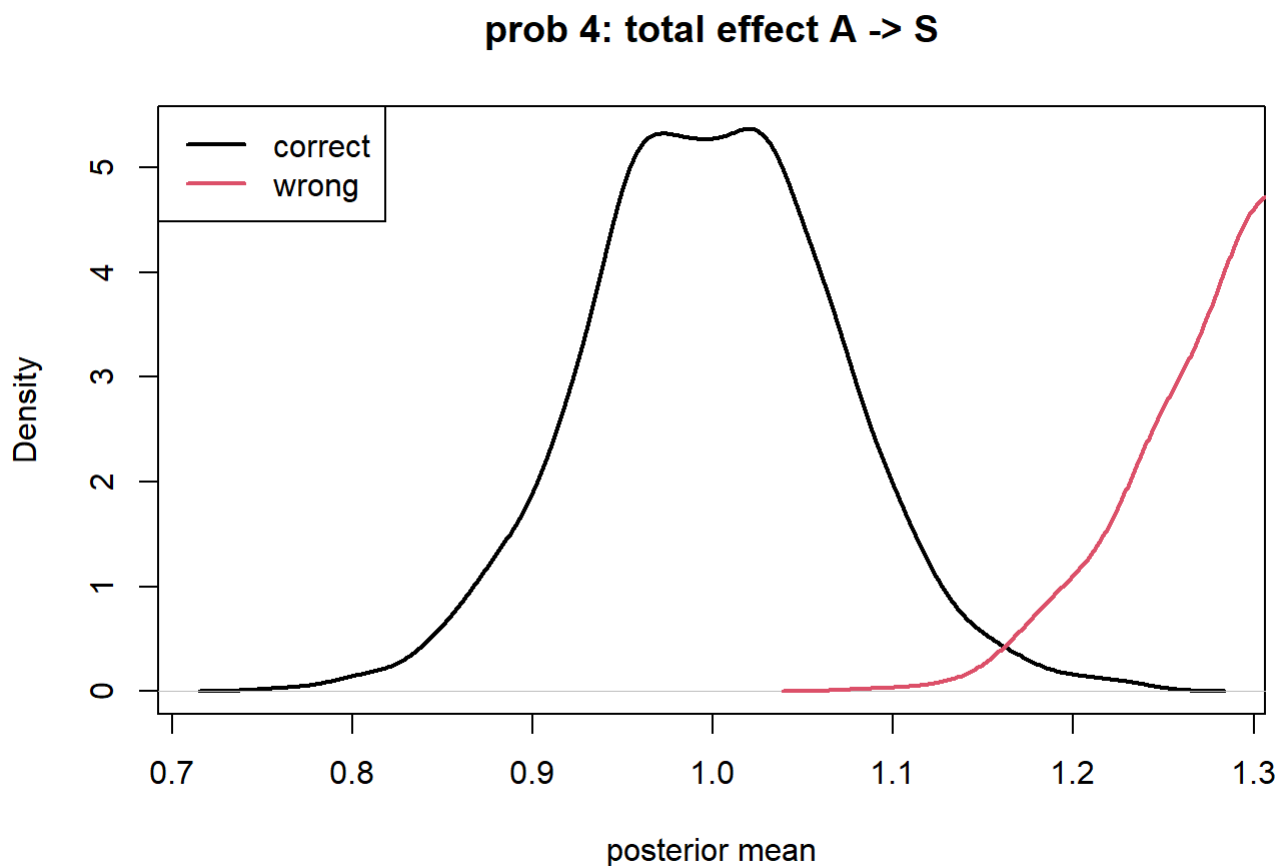
sim_p4 = function(n=100) {
  EC <- rnorm(n)
  B = rnorm(n, EC)
  Q <- rnorm(n)
  A = rnorm(n, B + Q)
  S <- rnorm(n, A + Q + EC)

  # correct: control Q and EC
  correct = coef(lm(S ~ A + Q + EC))['A']
  # wrong: forget to control Q
  wrong <- coef(lm(S ~ A + EC))['A']

  return(c(correct, wrong))
}

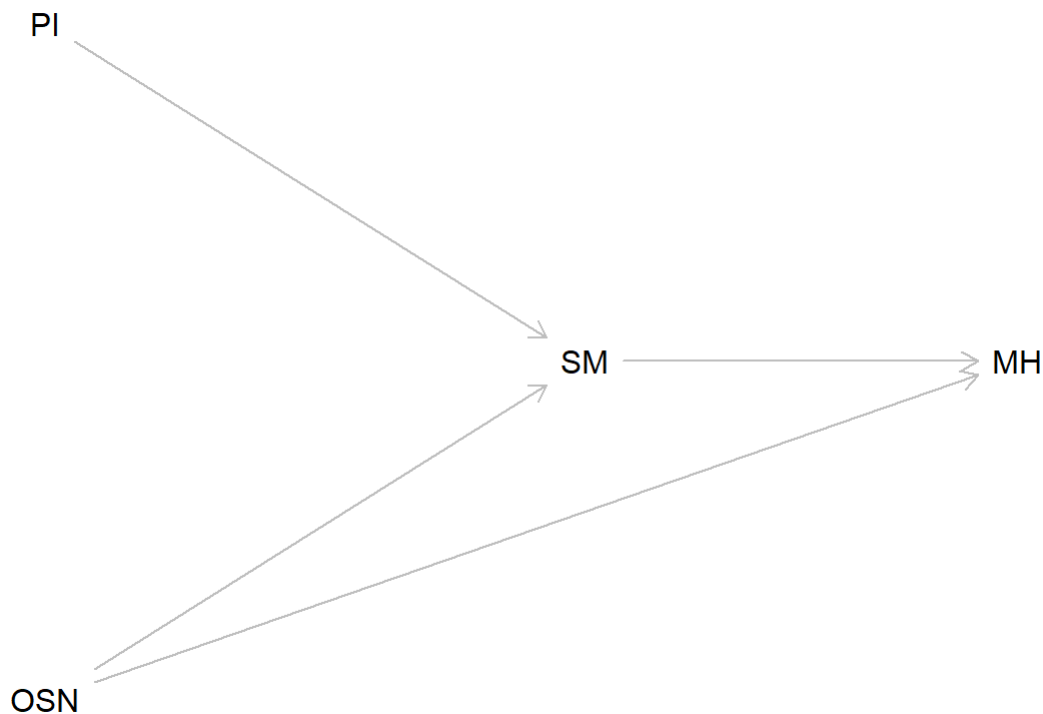
res4 = replicate(1000, sim_p4())
plot(density(res4[1,]), lwd=2, main="prob 4: total effect A -> S", xlab="posterior mean")
lines(density(res4[2,]), lwd=2, col=2)
legend("topleft", legend=c("correct", "wrong"), col=c(1, 2), lwd=2)

```



problem 5: social media use and mental health


```
p5_dag <- dagitty("dag {
  OSN -> SM
  OSN -> MH
  PI -> SM
  SM -> MH
}")
coordinates(p5_dag) <- list(
  x = c(OSN = 0, PI = 0, SM = 2, MH = 3.6),
  y = c(OSN = 1.5, PI = 0.5, SM = 1, MH = 1)
)
plot(p5_dag)
```



dag: OSN -> SM OSN -> MH PI -> SM SM -> MH

include in the model: OSN

do not include: PI

explanation: studying relationship between social media (SM) and mental health (MH). hours spent could directly affect mental health. offline network (OSN) influence both. personal interests (PI) lead to different SM use.

to get the direct effect we must control for OSN because it is a confounder.

part 2 wants us to show why we don't want to include personal interests in the model. PI only influences SM, so it works like an instrumental / precision parasite variable here. including it does not fix bias, because there is no backdoor through PI. it mostly makes the estimate less precise. so in the plot below the mean stays about the same, but the distribution with PI is wider.

```

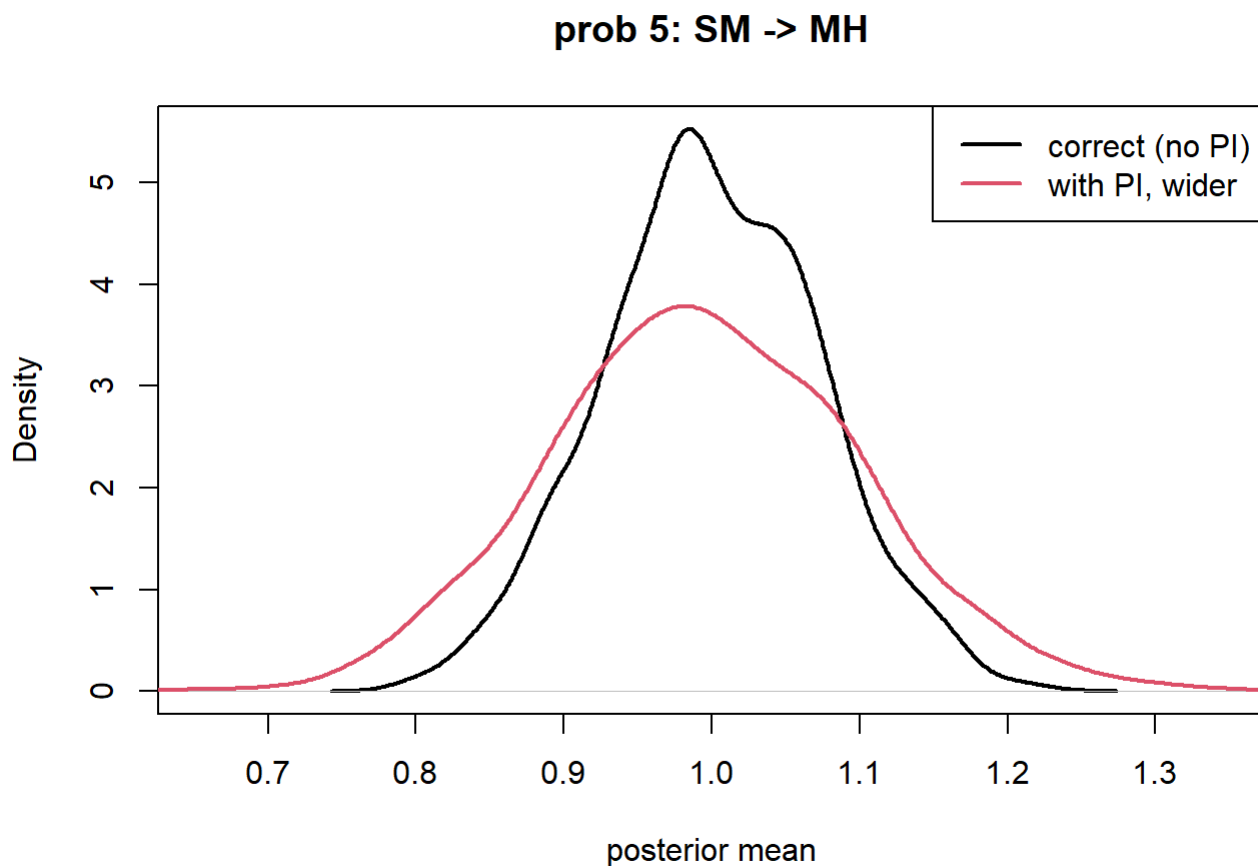
f5 <- function(n=100) {
  OSN = rnorm(n)
  PI = rnorm(n)
  SM <- rnorm(n, OSN + PI)
  MH <- rnorm(n, OSN + SM)

  # correct: control OSN only
  correct <- coef(lm(MH ~ SM + OSN))['SM']
  # not needed: adding PI mostly hurts precision
  wrong <- coef(lm(MH ~ SM + OSN + PI))['SM']

  return(c(correct, wrong))
}

sim5 <- replicate(1000, f5())
plot(density(sim5[1,]), lwd=2, main="prob 5: SM -> MH", xlab="posterior mean", xlim=range(sim5))
lines(density(sim5[2,]), lwd=2, col=2)
legend("topright", legend=c("correct (no PI)", "with PI, wider"), col=c(1, 2), lwd=2)

```



problem 6: pesticide exposure and health outcomes

```
p6_dag <- dagitty("dag {
D -> PE
D -> HC
PE -> HC
HC -> US
}")
coordinates(p6_dag) <- list(
  x = c(D = 0, PE = 1.5, HC = 3, US = 4.2),
  y = c(D = 1, PE = 1, HC = 1, US = 0)
)
plot(p6_dag)
```

US

(-0.5) DAG is wrong. Scenario says D influences both PE and HC, independently.

D —————> PE —————> HC

dag: D -> PE D -> HC PE -> HC HC -> US

include in the model: D

do not include: US

explanation: researchers studying effect of pesticide exposure (PE) on health condition (HC). diet (D) influence PE and HC. sunscreen (US) is used because HC makes skin sensitive.

to estimate the direct effect we have to control for diet (D) because it is a confounder. we must exclude sunscreen (US). US is a descendant of the outcome HC, so it is a post-treatment bad control and i dont want it in the model.

```

sim_p6 <- function(n=100) {
  D = rnorm(n)
  PE = rnorm(n, D)
  HC <- rnorm(n, PE + D)
  US <- rnorm(n, HC)

  # correct: control D, exclude US
  correct = coef(lm(HC ~ PE + D))['PE']
  # wrong: controlling descendant US
  wrong = coef(lm(HC ~ PE + D + US))['PE']

  return(c(correct, wrong))
}

res6 = replicate(1000, sim_p6())
plot(density(res6[1,]), lwd=2, main="prob 6: direct effect PE -> HC", xlab="posterior mean")
lines(density(res6[2,]), lwd=2, col=2)
legend("topright", legend=c("correct", "wrong (with US)"), col=c(1, 2), lwd=2)

```

prob 6: direct effect PE -> HC

